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--Project: Credit Risk Analysis

-- Objective:
--Analyze loan application data to identify factors that
-- contribute to borrower default risk.

-- Business Questions:

-- 1.What is the overall default rate, and how does it break down by credit score range (e.g., 520–599, 600–649, 650–699, 700–749, 750+)? Which credit score bucket has the highest default rate?

-- 2. Is there a relationship between a borrower's debt-to-income (DTI) ratio and the likelihood of defaulting? What DTI threshold would you recommend as a cutoff for loan approval?

--3. Which loan purposes have the highest default rates, and does the average loan amount differ significantly between defaulted and non-defaulted loans?

--4. How do employment status and years employed affect default risk? Are borrowers with less than 2 years of employment significantly more likely to default?
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--What is the overall default rate, and how does it break down by credit score range (e.g., 520–599, 600–649, 650–699, 700–749, 750+)?
--Which credit score bucket has the highest default rate?

SELECT
  Credit_score_range,Total_number_of_loans,
  Number_of_defaulted_loans,

  -- Calculate default rate for each credit score range
  CAST(
    (Number_of_defaulted_loans * 100.0) / Total_number_of_loans
    AS decimal(10,2)
  ) AS Default_rate_percentage

FROM (
  SELECT
    CASE
      WHEN b.credit_score BETWEEN 520 AND 599 THEN '520-599'
      WHEN b.credit_score BETWEEN 600 AND 649 THEN '600-649'
      WHEN b.credit_score BETWEEN 650 AND 699 THEN '650-699'
      WHEN b.credit_score BETWEEN 700 AND 749 THEN '700-749'
      WHEN b.credit_score >= 750 THEN '750+'
    END AS Credit_score_range,
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       COUNT(l.loan_amount) AS Total_number_of_loans,

 COUNT(CASE WHEN l.defaulted = 1 THEN 1 END) AS Number_of_defaulted_loans

    FROM borrower_profiles AS b
    JOIN loan_applications AS l
       ON b.borrower_id = l.borrower_id

    GROUP BY
       CASE
          WHEN b.credit_score BETWEEN 520 AND 599 THEN '520-599'
          WHEN b.credit_score BETWEEN 600 AND 649 THEN '600-649'
          WHEN b.credit_score BETWEEN 650 AND 699 THEN '650-699'
          WHEN b.credit_score BETWEEN 700 AND 749 THEN '700-749'
          WHEN b.credit_score >= 750 THEN '750+'
       END

) AS credit_score_summary;



---Is there a relationship between a borrower's debt-to-income (DTI) ratio and the likelihood of defaulting?
-- What DTI threshold would you recommend as a cutoff for loan approval?

WITH DTI_Data AS (
    SELECT
       CASE
          WHEN l.dti_ratio BETWEEN 0 AND 20 THEN '0% - 20%'
          WHEN l.dti_ratio BETWEEN 21 AND 30 THEN '21% - 30%'
          WHEN l.dti_ratio BETWEEN 31 AND 40 THEN '31% - 40%'
          WHEN l.dti_ratio BETWEEN 41 AND 50 THEN '41% - 50%'
          ELSE '51%+'
       END AS DTI_Range,

       COUNT(*) AS Total_number_of_loans,

       COUNT(CASE WHEN l.defaulted = 1 THEN 1 END) AS Number_of_defaulted_loans

    FROM loan_applications AS l
    JOIN borrower_profiles AS p
       ON l.borrower_id = p.borrower_id

    GROUP BY
       CASE
          WHEN l.dti_ratio BETWEEN 0 AND 20 THEN '0% - 20%'
          WHEN l.dti_ratio BETWEEN 21 AND 30 THEN '21% - 30%'
          WHEN l.dti_ratio BETWEEN 31 AND 40 THEN '31% - 40%'
          WHEN l.dti_ratio BETWEEN 41 AND 50 THEN '41% - 50%'
      ELSE '51%+'
       END
)
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SELECT
  DTI_Range,
  Total_number_of_loans,
  Number_of_defaulted_loans,

  CAST(
    (Number_of_defaulted_loans * 100.0) / Total_number_of_loans
    AS DECIMAL(10,2)
  ) AS Default_rate_percentage

FROM DTI_Data;



--- Which loan purposes have the highest default rates, and does the average loan amount differ significantly between defaulted and non-defaulted loans?
SELECT
  l.loan_purpose,

  COUNT(*) AS Total_number_of_loans,

  COUNT(CASE WHEN l.defaulted = 1 THEN 1 END) AS Number_of_defaulted_loans,

  CAST(
    COUNT(CASE WHEN l.defaulted = 1 THEN 1 END) * 100.0 / COUNT(*)
    AS decimal(10,2)
  ) AS Default_rate_percentage,

  CAST(
    AVG(CASE WHEN l.defaulted = 1 THEN l.loan_amount END)
    AS decimal(10,2)
  ) AS Average_Defaulted_Loan_Amount,

  CAST(
    AVG(CASE WHEN l.defaulted = 0 THEN l.loan_amount END)
    AS decimal(10,2)
  ) AS Average_Non_Defaulted_Loan_Amount

FROM loan_applications AS l

JOIN borrower_profiles AS p
  ON l.borrower_id = p.borrower_id

GROUP BY l.loan_purpose

ORDER BY Default_rate_percentage DESC;



-- Compare average loan amount between defaulted and non-defaulted loans

SELECT
  CASE
    WHEN l.defaulted = 1 THEN 'Defaulted'
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        ELSE 'Non-Defaulted'
    END AS Loan_Status,

    CAST(
        ROUND(AVG(l.loan_amount), 2)
        AS decimal(10,2)
    ) AS Average_loan_amount

FROM loan_applications AS l
JOIN borrower_profiles AS p
    ON l.borrower_id = p.borrower_id

GROUP BY l.defaulted

ORDER BY l.defaulted;
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---How do employment status and years employed affect default risk?
--Are borrowers with less than 2 years of employment significantly more likely to default?
SELECT
    p.employment_status,

    CASE
        WHEN p.years_employed < 2 THEN 'Less than 2 years'
        ELSE '2 years or more'
    END AS Employment_Group,

    COUNT(*) AS Total_Loans,

    COUNT(CASE WHEN l.defaulted = 1 THEN 1 END) AS Defaulted_Loans,

    CAST(
        COUNT(CASE WHEN l.defaulted = 1 THEN 1 END) * 100.0 / COUNT(*)
        AS DECIMAL(10,2)
    ) AS Default_Rate_Percentage

FROM loan_applications AS l
JOIN borrower_profiles AS p
    ON l.borrower_id = p.borrower_id

GROUP BY
    p.employment_status,

    CASE
        WHEN p.years_employed < 2 THEN 'Less than 2 years'
        ELSE '2 years or more'
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    END

ORDER BY
    Default_Rate_Percentage DESC;

    -- income  buckets
SELECT
    INCOME_RANGES,
    Total_number_of_loans,
    CONCAT('$', FORMAT(CAST(Average_Loan_Amount AS DECIMAL(10,2)), 'N2')) AS Average_Loan_Amount,
    Number_of_defaulted_loans,
    CAST(
        (Number_of_defaulted_loans * 100.0) / Total_number_of_loans
        AS DECIMAL(10,2)
    ) AS Default_rate_percentage

FROM (
    SELECT
        CASE
            WHEN b.annual_income < 20000 THEN '$19,999 or less'
            WHEN b.annual_income BETWEEN 20000 AND 30000 THEN '$20,000-$30,000'
            WHEN b.annual_income BETWEEN 30001 AND 40000 THEN '$30,001-$40,000'
            WHEN b.annual_income BETWEEN 40001 AND 50000 THEN '$40,001-$50,000'
            WHEN b.annual_income BETWEEN 50001 AND 60000 THEN '$50,001-$60,000'
            WHEN b.annual_income BETWEEN 60001 AND 70000 THEN '$60,001-$70,000'
            WHEN b.annual_income BETWEEN 70001 AND 80000 THEN '$70,001-$80,000'
            WHEN b.annual_income BETWEEN 80001 AND 90000 THEN '$80,001-$90,000'
            WHEN b.annual_income BETWEEN 90001 AND 100000 THEN '$90,001-$100,000'
            ELSE '$100,001+'
        END AS INCOME_RANGES,

        COUNT(*) AS Total_number_of_loans,

        AVG(CAST(l.loan_amount AS DECIMAL(10,2))) AS Average_Loan_Amount,

        COUNT(CASE WHEN l.defaulted = 1 THEN 1 END) AS Number_of_defaulted_loans

    FROM loan_applications AS l
    JOIN borrower_profiles AS b
        ON l.borrower_id = b.borrower_id

    GROUP BY
        CASE
            WHEN b.annual_income < 20000 THEN '$19,999 or less'
            WHEN b.annual_income BETWEEN 20000 AND 30000 THEN '$20,000-$30,000'
            WHEN b.annual_income BETWEEN 30001 AND 40000 THEN '$30,001-$40,000'
            WHEN b.annual_income BETWEEN 40001 AND 50000 THEN '$40,001-$50,000'
            WHEN b.annual_income BETWEEN 50001 AND 60000 THEN '$50,001-$60,000'
            WHEN b.annual_income BETWEEN 60001 AND 70000 THEN '$60,001-$70,000'
            WHEN b.annual_income BETWEEN 70001 AND 80000 THEN '$70,001-$80,000'
            WHEN b.annual_income BETWEEN 80001 AND 90000 THEN '$80,001-$90,000'
            WHEN b.annual_income BETWEEN 90001 AND 100000 THEN '$90,001-$100,000'
            ELSE '$100,001+'
        END
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) AS Income_Table

ORDER BY
  CASE
    WHEN INCOME_RANGES = '$19,999 or less' THEN 1
    WHEN INCOME_RANGES = '$20,000-$30,000' THEN 2
    WHEN INCOME_RANGES = '$30,001-$40,000' THEN 3
    WHEN INCOME_RANGES = '$40,001-$50,000' THEN 4
    WHEN INCOME_RANGES = '$50,001-$60,000' THEN 5
    WHEN INCOME_RANGES = '$60,001-$70,000' THEN 6
    WHEN INCOME_RANGES = '$70,001-$80,000' THEN 7
    WHEN INCOME_RANGES = '$80,001-$90,000' THEN 8
    WHEN INCOME_RANGES = '$90,001-$100,000' THEN 9
    WHEN INCOME_RANGES = '$100,001+' THEN 10
  END;
```